

BHAVANI DAIRY FARM

Complete Feed Preparation, Health & Cost Management Guide

40 Murrah
Buffaloes

Herd size

Pydi
Bhimavaram,
Srikakulam, AP

Location

NABARD DEDS
■44.7L Subsidy

Scheme

2026 Prices &
Protocols

Reference year

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SECTION 1 — FEED INGREDIENTS: LOCAL AVAILABILITY & 2026 PRICES

Srikakulam district is a major paddy and sugarcane belt with good availability of green fodder, paddy by-products and oilseed cakes. The following ingredients are all procurable within 30–50 km of Pydi Bhimavaram.

1.1 Green Fodder

Ingredient	Season Availability	Source	Price/kg (₹)	Dry Matter %	Notes
Napier Grass (CO-3/CO-4)	Year-round	Own cultivation / local market	0.50–0.80	20%	Best year-round green; plant once, harvest 6–8x/year
Maize (green)	Kharif (Jul–Oct) Rabi (Jan–Mar)	Local farmers	1.20–1.80	25%	High energy; ideal for silage
Sorghum / Jowar	Kharif & Rabi	Local markets	0.80–1.20	22%	Good palatability; drought tolerant
Subabul (Leucaena)	Year-round	Bund planting	0	30%	Free protein bank; plant on farm bunds
Para Grass	Year-round	Canal banks	0–0.50	18%	Available near water channels in Srikakulam
Sugarcane tops	Nov–Feb	Local sugarcane farmers	0.20–0.40	24%	Seasonal; low cost, bulk filler
Sweet Sorghum bagasse	Nov–Jan	Sugar mills	0.15–0.30	28%	Energy-rich by-product

1.2 Dry Fodder

Ingredient	Availability	Price/kg (₹)	DM %	Crude Protein %	Notes
Paddy Straw	Year-round (abundant)	0.50–1.00	88 %	3–4%	Most abundant; low nutrition — treat with urea
Urea-treated Straw	Year-round	1.00–1.50	88 %	8–10%	Treat paddy straw with 4% urea solution — triples protein
Maize Stover	Post-harvest	0.80–1.20	86 %	6%	Better than paddy straw; good energy

Ingredient	Availability	Price/kg (₹)	DM %	Crude Protein %	Notes
Groundnut Haulms	Nov–Jan	2.00–3.00	90 %	12%	Excellent protein in dry fodder — stock up in season
Soybean Stover	Oct–Nov	1.50–2.00	88 %	8%	Available from Vizianagaram belt

1.3 Concentrate Ingredients

Ingredient	Price/kg (₹)	Crude Protein %	TDN %	Source in Srikakulam
Cotton Seed Cake (expeller)	22–26	38–41%	72%	Dealer: Srikakulam / Palasa road
Groundnut Cake (expeller)	28–34	45–48%	74%	Available in Amadalavalasa, Narasannapeta
Rice Bran (deoiled)	14–18	12–14%	68%	Rice mills in Narasannapeta, Srikakulam town
Maize Grain	20–24	9%	80%	APMC Srikakulam / Etcherla market
Wheat Bran	16–20	14%	65%	Available from Vizag / Srikakulam wholesalers
Broken Rice	18–22	8%	78%	Rice mills — abundant and cheap locally
Soybean Meal (solvent)	35–40	44–46%	72%	Vizag-based dealers supply to Srikakulam
Molasses	8–12	3%	60%	Sugar factories in Chipurupalle area

1.4 Minerals & Supplements

Supplement	Price/kg (₹)	Use	Source
Mineral Mixture (ISI-marked)	55–75	50 g/animal/day	Virbac, Provimi, Godrej Agrovet — any feed dealer
Common Salt (Iodised)	8–12	30 g/animal/day	Any grocery / feed store
Limestone Powder	3–5	50 g/animal/day	Building material suppliers
Bypass Fat (Megalac)	120–140	100–150 g/day (peak)	Dairy dealers; Vizag
Sodium Bicarbonate	25–30	100 g/day (summer)	Chemical suppliers, Vizag

■ **Tip:** Join the Srikakulam District Milk Producers Co-operative Society (SDMPCS) to access subsidised concentrate feed, mineral mixture and veterinary services at reduced cost.

SECTION 2 — DAILY RATION DESIGN FOR MURRAH BUFFALO

Based on ICAR-NDRI and NDDB guidelines for high-yielding Murrah buffaloes under tropical conditions (Andhra Pradesh). Requirements vary by physiological stage.

2.1 Nutrient Requirements by Stage

Stage	Body Wt (kg)	Milk (L/day)	Dry Matter (kg/day)	Crude Protein (g/day)	TDN (kg/day)	Calcium (g)	Phosphorus (g)
Maintenance only	450	0	6.0	480	3.2	18	14
Early lactation (8–12 L)	450	10	13.5	1,450	8.1	55	40
Peak lactation (12–16 L)	450	14	16.0	1,750	9.6	65	48
Late lactation (6–8 L)	450	7	11.0	1,100	6.5	42	30
Dry period (2 months)	470	0	8.5	900	5.2	40	28
Last 30 days pregnancy	490	0	9.5	1,100	6.0	55	38

2.2 Recommended Daily Balanced Ration (per animal, 14 L/day peak lactation)

This is the standard ration recommended for your 31 lactating Murrah buffaloes producing ~14 litres/day:

Feed Ingredient	Qty (kg /day)	DM (kg)	CP (g)	TDN (kg)	Cost/day (₹)
Green fodder (Napier/Maize)	25	5.0	425	3.0	15.00
Urea-treated paddy straw	5	4.4	440	2.5	6.25
Maize grain	2	1.76	180	1.6	44.00
Rice bran (deoiled)	1.5	1.35	195	1.0	24.00
Cotton seed cake	1.5	1.38	585	1.1	36.00
Groundnut cake	0.5	0.46	225	0.37	15.00
Molasses	0.3	0.24	9	0.18	3.00

Feed Ingredient	Qty (kg /day)	DM (kg)	CP (g)	TDN (kg)	Cost/day (₹)
Mineral mixture	0.05	0.05	—	—	3.25
Common salt	0.03	0.03	—	—	0.30
Limestone powder	0.05	0.05	—	—	0.25
TOTAL	35.93	14.72	2,059	9.75	₹147.05

■ This ration meets ICAR requirements: DM=14.7 kg (target 14–16 kg), CP=2,059 g (target 1,750 g), TDN=9.75 kg (target 9.6 kg), Ca:P ratio = 1.4:1 (ideal). Daily cost per peak-lactating animal = ■147.

For **dry animals** (9 animals): reduce concentrate to 2 kg/day total, maintain green + straw — cost drops to ~■80/day.

SECTION 3 — THREE FEEDING MODELS COMPARED

Three practical models suited to Srikakulam conditions. All are designed for 40 Murrah buffaloes (31 lactating, 9 dry/pregnant).

MODEL A — Fully Purchased Feed (Zero Own Cultivation)

All green fodder, dry fodder and concentrate purchased from market. Suitable if land is not available.

Item	Qty/day (40 animals)	Unit Price (₹)	Daily Cost (₹)	Monthly Cost (₹)
Green fodder (purchased Napier)	600 kg	0.70/kg	420	12,600
Paddy straw (purchased)	120 kg	1.00/kg	120	3,600
Maize grain	56 kg	22/kg	1,232	36,960
Rice bran	42 kg	16/kg	672	20,160
Cotton seed cake	42 kg	24/kg	1,008	30,240
Groundnut cake	14 kg	30/kg	420	12,600
Molasses	8 kg	10/kg	80	2,400
Minerals + salt	3.2 kg	60/kg	192	5,760
MODEL A TOTAL			₹4,144	₹1,24,320

■■ **Cost: ₹1,24,320/month.** Highest cost model. Risk: green fodder availability varies seasonally. Not recommended as primary model.

MODEL B — Mixed Model (Own Napier + Purchased Concentrate) ★ RECOMMENDED

Cultivate 2.5 acres of Napier/Maize for green fodder. Purchase all concentrate. This is the optimal balance for a 40-buffalo farm in Srikakulam.

Item	Qty/day	Cost basis	Daily Cost (₹)	Monthly Cost (₹)
Own Napier grass (2.5 ac)	500 kg/day	Own cultivation	125	3,750
Supplemental green (purchased)	100 kg/day	₹0.70/kg	70	2,100
Urea-treated paddy straw (own)	120 kg/day	₹0.40/kg processed	48	1,440
Maize grain	56 kg/day	₹22/kg	1,232	36,960

Item	Qty/day	Cost basis	Daily Cost (₹)	Monthly Cost (₹)
Rice bran	42 kg/day	₹16/kg	672	20,160
Cotton seed cake	42 kg/day	₹24/kg	1,008	30,240
Groundnut cake	14 kg/day	₹30/kg	420	12,600
Minerals + salt	3.2 kg/day	₹60/kg	192	5,760
MODEL B TOTAL			₹3,767	₹1,13,010

■ **Cost: ₹1,13,010/month.** Saves ₹11,310/month vs Model A. 2.5 acres of Napier requires one-time planting cost of ~₹35,000 but lasts 5+ years with minimal maintenance.

MODEL C — Silage-Based Model (Maize Silage + Concentrate)

Make maize or sorghum silage during Kharif season (Aug–Oct) for year-round assured green fodder supply. Requires 3 acres and silage pit infrastructure.

Item	Details	Annual Cost (₹)	Monthly equiv. (₹)
Maize silage making (3 ac)	60 T silage/yr — 550 kg/day	75,000	6,250
Silage pit (one-time, 5yr life)	60T pit — concrete	80,000 ÷ 5 = 16,000/yr	1,333
Concentrate (reduced 20%)	Lower qty vs B — silage covers energy	10,10,000	84,167
Minerals + salt	Same as Model B	69,120	5,760
MODEL C TOTAL		₹11,70,120	₹97,510

■■ **Cost: ₹97,510/month.** Lowest operating cost but requires upfront silage pit investment (~₹80,000). Best for farms with 3+ acres land. Seasonal maize availability in Srikakulam makes this very practical.

3.4 Model Comparison Summary

Model	Monthly Feed Cost	Land Needed	Upfront Investment	Seasonal Risk	Verdict
A — Full Purchase	₹1,24,320	None	Nil	High	Not recommended
B — Own Napier + Purchase	₹1,13,010	2.5 acres	₹35,000	Low	★ Best for your farm
C — Silage-based	₹97,510	3 acres	₹80,000	Very Low	Best if 3 ac available

SECTION 4 — MONTHLY FEED COST BREAKDOWN (40 BUFFALOES, MODEL B)

Detailed monthly feed cost for Model B — the recommended model. Costs vary slightly by season due to green fodder availability.

Month	Season	Green Fodder Cost	Dry Fodder Cost	Concentrate Cost	Minerals	Total/Month (₹)
January	Winter/Rabi	4,500	2,500	1,00,000	5,760	1,12,760
February	Winter/Rabi	4,500	2,500	1,00,000	5,760	1,12,760
March	Summer	5,500	3,000	1,05,000	5,760	1,19,260
April	Summer (hot)	6,500	3,500	1,05,000	6,500	1,21,500
May	Summer (peak)	7,000	4,000	1,10,000	6,500	1,27,500
June	Pre-monsoon	5,500	3,000	1,05,000	5,760	1,19,260
July	Monsoon	3,750	1,440	1,00,960	5,760	1,11,910
August	Monsoon	3,750	1,440	1,00,960	5,760	1,11,910
September	Monsoon/Kharif	3,750	1,440	1,00,960	5,760	1,11,910
October	Post-monsoon	3,750	1,440	1,00,960	5,760	1,11,910
November	Cool/Rabi	4,000	1,800	1,00,960	5,760	1,12,520
December	Winter	4,500	2,500	1,00,000	5,760	1,12,760
ANNUAL TOTAL		₹57,000	₹28,560	₹12,29,800	₹70,080	₹13,56,960

■ **Annual feed cost (Model B): ₹13,56,960 (~₹13.57 lakhs)** · Average **₹1,13,080/month** · Cost per litre of milk (at 426 L/day): **≈ ₹8.74/litre**

Summer months (March–May) cost is higher due to: purchased green fodder, extra sodium bicarbonate for heat stress, and electrolyte supplements. Plan extra budget of **₹5,000–8,000/month** for May.

SECTION 5 — SEASONAL MEDICATION & VACCINATION CALENDAR

Schedule based on Andhra Pradesh Animal Husbandry Department guidelines and ICAR recommendations for coastal AP conditions. Some vaccines are provided FREE by the AP government through the local VAS (Veterinary Assistant Surgeon).

5.1 Annual Vaccination Calendar

Month	Disease	Vaccine / Drug	Dose	Route	Cost/animal (₹)	Provided By
January	Brucellosis (calves 4–8 mo)	Brucella abortus S19	2 mL	SC	Free	Govt. VAS
February	FMD	FMD Polyvalent (O, A, Asia-1)	2 mL	SC	Free	Govt. — biannual drive
February	Deworming	Albendazole 10% bolus	1 bolus (750 mg)	Oral	15–20	Purchase
April	Theileriosis (calves)	Theileria annulata schizonts	1 mL	SC	80–100	SVBP / Pvt. vet
May	Heat stress support	Electral / ORS + Vit E+Se	As per wt	Oral	20–30	Purchase
June	HS (Haemorrhagic Septicaemia)	HS Alum precipitate vaccine	2 mL	SC	Free	Govt. — pre-monsoon
June	BQ (Black Quarter)	BQ vaccine (spore suspension)	5 mL	SC	Free	Govt. — pre-monsoon
August	Deworming	Albendazole or Fenbendazole	750 mg	Oral	15–20	Purchase
August	FMD (second dose)	FMD Polyvalent	2 mL	SC	Free	Govt. — biannual drive
September	Anti-tick treatment	Deltamethrin / Cypermethrin spray	Dilute & spray	Topical	10–15	Purchase
October	Liver tonic + Vit B-complex	Livol / Liv-52 Vet oral	50 mL/day × 10d	Oral	30–40	Purchase
November	Deworming	Ivermectin pour-on or bolus	200 mcg/kg	Pour-on	40–50	Purchase

Month	Disease	Vaccine / Drug	Dose	Route	Cost/animal (₹)	Provided By
December	Vitamin AD3E injection	AD3E inj (Vit Forte)	5 mL	IM	20–25	Purchase

SC = Subcutaneous · IM = Intramuscular · Govt. VAS = free through AP Animal Husbandry veterinarian visiting your village

5.2 Seasonal Health Interventions

Season	Challenge	Intervention	Product	Cost/month (40 animals)
Summer (Mar–Jun)	Heat stress, reduced milk, reduced appetite	- Sodium bicarbonate 100g/day - Electrolytes in water - Vit E + Selenium injection (Apr) - Increase water to 80L/animal/day - Shade + fans mandatory	Electral, Bicarb, Vit E-Se	■ 3,500–4,500
Monsoon (Jul–Sep)	Foot rot, FMD, HS, mastitis, worm load spike	- Pre-monsoon HS + BQ vaccination (June) - Deworming in August - Foot bath (5% formalin) twice/week - Tick control spray monthly - Watch for laminitis in wet conditions	Formalin, Cypermethrin, Albendazole	■ 2,000–3,000
Winter (Oct–Feb)	Reduced water intake, milk fever risk, mastitis in fresh cows	- Liver tonic course (October) - Vit AD3E injection (December) - Calcium bolus for fresh animals - Deworming (November) - Watch for cold stress in calves	Livol, Vit Forte, Calcigel bolus	■ 1,500–2,500

SECTION 6 — MINERALS, VITAMINS & SUPPLEMENTS PROTOCOL

Srikakulam district has coastal alluvial soils that are commonly deficient in selenium, zinc, copper and iodine. Murrah buffaloes are especially sensitive to selenium and zinc deficiency (impacts milk fat, reproduction and hoof health).

6.1 Daily Mineral Supplementation

Mineral/Supplement	Daily Dose	Function	Deficiency Signs	Annual Cost/animal (₹)
Mineral mixture (ISI-marked)	50 g/day	Multi-mineral: Ca, P, Mg, Zn, Cu, Mn, Se, I, Co	Reduced milk, poor reproduction, weak bones	900–1,100
Common salt (iodised)	30 g/day	Electrolyte balance, iodine supplementation	Reduced appetite, goitre, poor calving	90–120
Limestone powder	50 g/day	Calcium supplementation (low-cost)	Milk fever, fragile bones, poor milk yield	55–70
Bypass fat (Megalac)	100 g/day (lactating only)	Energy density, early lactation support	Excessive weight loss, ketosis	1,800–2,000
Sodium bicarbonate	100 g/day (summer only)	Rumen buffering, heat stress relief	Acidosis, off-feed in summer	100–130

6.2 Targeted Injectable Supplements (Twice Per Year)

Injection	When	Dose	Why Important for Srikakulam	Cost/dose (₹)
Selenium + Vit E (Selevit / E-Se)	April & October	5 mL IM	Coastal soils are Se-deficient; prevents white muscle disease, improves conception rate	40–60
Vitamin AD3E (Vit Forte)	December & June	5 mL IM	Supports immunity, fat-soluble vitamin replenishment post-monsoon	20–30
Vitamin B-complex	After deworming (Aug, Nov)	10 mL IM	Supports recovery, appetite stimulation post-anthelmintic	15–20
Calcium borogluconate (IV)	At calving (as needed)	400 mL IV slow	Milk fever prevention — critical in high-yielding buffaloes	80–120

6.3 Annual Medicine & Supplement Cost Per Animal

Category	Annual Cost/Animal (■)	Cost for 40 Animals (■)
Mineral mixture (daily)	1,000	40,000
Salt + Limestone (daily)	190	7,600
Bypass fat (lactating)	1,900	58,900
Vaccines (purchased)	220	8,800
Deworming (3x/year)	150	6,000
Injectable supplements	250	10,000
Tick/ectoparasite control	120	4,800
Summer electrolytes + bicarb	350	14,000
Emergency/sick animal medicines	500	20,000
TOTAL ANNUAL	■4,680	■1,70,100

■ Many vaccines (FMD, HS, BQ) are provided FREE by AP Animal Husbandry Dept. Register with your local VAS office in Pydi Bhimavaram to get these services. Also eligible under AP NTR Pashu Seva scheme.

SECTION 7 — ANIMAL HEALTH TRACKING SYSTEM

A systematic daily health observation and recording system reduces disease losses by 30–40%. Use the DairyFlow app (already set up on your farm) or this paper-based protocol.

7.1 Daily Observation Checklist (Morning Shift)

Parameter	Normal Range	Action if Abnormal
Body temperature	101–103°F (38.3–39.4°C)	>104°F: isolate, call vet; <100°F: check for ketosis/milk fever
Rumination	6–8 hours/day, ~60 chews/bolus	No rumination for >12h: off-feed, rumen issue — consult vet
Milk yield	Within 10% of previous day	>15% drop: check for mastitis, fever, stress
Milk appearance	Uniform, white, no clots	Clots/watery/blood: CMT test immediately for mastitis
Appetite	Finishes feed in 30 min	Leaving feed: fever, acidosis, off-feed — investigate
Dung consistency	Firm cowpat shape	Watery: dietary change / infection; Hard: dehydration / blockage
Udder condition	Soft, no heat/swelling	Hard quarter / heat: mastitis — CMT, antibiotic as per vet
Lameness / gait	Normal walking	Limp: foot rot, laminitis — clean foot, trim, treat
Nasal / eye discharge	None or minimal clear	Thick discharge: respiratory infection — isolate, treat
Reproductive signs	Record heat (restlessness, mucus)	AI within 12–18h of standing heat for best conception

7.2 Reproductive Events Tracking

Event	What to Record	Target / Ideal
Heat detection	Date, time, duration, intensity (standing heat)	Every 21 days (19–23 day cycle)
Artificial Insemination (AI)	Date, bull semen (tag no.), technician name, AI no. (1st/2nd/3rd)	12–18h after standing heat observed
Pregnancy diagnosis	Date, result (+ / –), method (rectal/ultrasound)	60–90 days after AI
Expected calving date	AI date + 305 days = expected calving	Gestation: 300–320 days

Event	What to Record	Target / Ideal
Calving record	Date, calf sex/weight, ease of calving, colostrum given	Calving interval target: 13–14 months
Dry-off	Date, milk yield at dry-off, dry cow therapy given	60 days before expected calving
Lactation record	Total milk per lactation, peak yield, lactation days	Target: >2,000 L per lactation

7.3 Common Diseases in Coastal AP Buffaloes — Quick Reference

Disease	Season	Signs	Treatment	Prevention
Mastitis	Year-round (peak summer)	Reduced milk, swollen quarter, clots	Intramammary AB (Amoxycyclav); systemic if severe	Post-milking teat dip (0.5% iodine), dry cow therapy
FMD	Monsoon (Jul–Sep)	Blisters on feet/mouth, high fever, milk drop	Supportive; wash blisters with KMnO ₄ solution	6-monthly FMD vaccine (Govt. free)
Haemorrhagic Septicaemia	Pre-monsoon	High fever (107°F), swollen neck, sudden death	Oxytetracycline IV immediately; Sulphonamides	Annual HS vaccine June (Govt. free)
Milk Fever	Post-calving	Unable to stand, cold extremities, low temp	Calcium borogluconate 400 mL IV slow — URGENT	Calcium bolus 48h + 12h before calving
Ketosis	Early lactation	Off-feed, sweet breath, weight loss	Glucose IV, propylene glycol oral, Vit B12	Body condition score (BCS) 3.0–3.5 at calving
Foot Rot	Monsoon	Foul smell, swollen foot, lameness	Oxytetracycline IM; foot bath 5% formalin	Dry standing areas, zinc supplementation
Theileriosis	Summer (vector ticks)	High fever, anaemia, swollen lymph nodes	Buparvaquone 2.5 mg/kg IM — URGENT	Theileria vaccine calves; tick control

SECTION 8 — WATER QUALITY TESTING

Water quality directly impacts buffalo health, milk yield and milk quality. Srikakulam coastal areas can have issues with fluoride, salinity and nitrate contamination in bore wells.

8.1 Water Consumption per Buffalo

Stage	Normal (L/day)	Summer (L/day)	For 40 Animals
Lactating (peak)	60–80	90–100	2,400–3,200 L/day
Dry / Pregnant	40–50	60–70	360–450 L/day
Total farm (40 animals)	~2,500 L	~3,500 L	Need 3,500 L/day peak capacity

8.2 Water Quality Parameters to Test

Parameter	Acceptable Limit (BIS/ BIS-11905)	Buffalo-S pecific Limit	Risk if Exceeded
pH	6.5–8.5	6.5–8.5	Acidic: rumen upset; Alkaline: mineral binding
Total Dissolved Solids (TDS)	<500 mg/L (ideal)	<1,000 mg/L	>3,000 mg/L: reduced intake, diarrhoea
Hardness (as CaCO ₃)	<300 mg/L	<500 mg/L	Excess Ca/Mg: urinary stones
Nitrates (NO ₃)	<45 mg/L	<100 mg/L	>200 mg/L: abortion, methemoglobinemia
Fluoride (F)	<1 mg/L	<2 mg/L	>4 mg/L: dental/bone fluorosis — common in Srikakulam bore wells
Chloride (Cl)	<250 mg/L	<600 mg/L	>1,000 mg/L: diarrhoea, reduced intake
Sulphates (SO ₄)	<200 mg/L	<500 mg/L	Laxative effect; reduced copper absorption
Total Coliform (E.coli)	Absent	Absent	Mastitis, diarrhoea, abortion
Iron (Fe)	<0.3 mg/L	<1 mg/L	Staining; affects palatability
Arsenic (As)	<0.01 mg/L	<0.05 mg/L	Chronic toxicity, poor production

8.3 Where to Test Water Near Srikakulam

Lab / Agency	Location	Tests Offered	Approx. Cost
APSPCB (AP Pollution Control Board)	Srikakulam District Office	Full chemical + bacteriological	■500–1,200
AP Water Technology Centre (APTC)	Vizag (Hyderabad HQ)	Complete water analysis	■800–1,500
ANGRAU Soil Testing Lab	Anakapalli / Vizianagaram	Chemical parameters	■400–800
Primary Health Centre (PHC) Lab	Pydi Bhimavaram / Sompeta	Bacteriological (coliform only)	■100–200
Private labs (Thyrocare / SRL)	Vizag collection centres	Chemical + bacteriological	■1,000–2,000

■ **Recommended frequency:** Test drinking water **once per year** (preferably post-monsoon, October–November) and after any new bore well is drilled. If fluoride or TDS comes high — install a RO system for animal drinking water.

SECTION 9 — SOIL TESTING FOR FODDER CULTIVATION

Srikakulam district has predominantly coastal alluvial soils (sandy loam to clay loam). These soils are generally fertile but can be acidic in upland areas and have micronutrient deficiencies common to the coastal belt.

9.1 Soil Tests Required Before Fodder Cultivation

Test	Parameter	Optimal Range (for fodder)	Why It Matters
Soil pH	Acidity / Alkalinity	6.0–7.5	Determines nutrient availability; below 5.5 — apply lime
Organic Carbon (OC)	Soil organic matter %	>0.75%	Indicates fertility; low OC → apply FYM/compost
Available Nitrogen (N)	kg N per hectare	>280 kg/ha (medium)	Critical for Napier / maize growth
Available Phosphorus (P)	kg P ₂ O ₅ per hectare	>25 kg/ha	Root development; common deficiency in coastal AP
Available Potassium (K)	kg K ₂ O per hectare	>280 kg/ha	Disease resistance; coastal soils often K-deficient
Sulphur (S)	Available sulphur ppm	>10 ppm	Protein synthesis in fodder crops
Zinc (Zn)	Available Zn ppm	>0.6 ppm	Very common deficiency in AP — causes poor germination
Boron (B)	Available B ppm	>0.5 ppm	Cell wall formation; deficiency in light soils
Soil EC	Electrical Conductivity (dS/m)	<1.0 dS/m	Salinity check — coastal farms may have saline seepage
Soil texture	Sand/Silt/Clay %	Sandy loam ideal	Drainage planning; Napier needs well-drained soil

9.2 Fertiliser Recommendations for Napier Grass (per acre)

Application	Fertiliser	Qty/acre	When	Cost/acre (₹)
Basal (planting)	FYM (Farm Yard Manure)	5 tonnes	15 days before planting	Use own — free
Basal	DAP (18:46:0)	50 kg	At planting	1,300

Application	Fertiliser	Qty/acre	When	Cost/acre (₹)
Basal	MOP (Muriate of Potash)	25 kg	At planting	575
Top dressing (after each cut)	Urea (46% N)	20 kg	After each harvest (~45 days)	440
Micronutrient (if deficient)	Zinc Sulphate	10 kg	Once per year	250
Total annual input cost				~₹3,500–4,500/acre

9.3 Where to Get Soil Tested Near Pydi Bhimavaram

Lab / Agency	Location	Tests	Cost	Turnaround
ANGRAU Soil Testing Lab	Vizianagaram campus (60 km from Pydi Bhimavaram)	N, P, K, pH, OC, EC, micronutrients	₹50–150	7–10 days
Dept. of Agriculture STL	Srikakulam District HQ	N, P, K, pH, OC, EC	Free (govt. scheme)	15–20 days
IFFCO / Coromandel dealer	Srikakulam town / Narasannapeta	Basic NPK + pH	Free with fertiliser purchase	5–7 days
Rythu Seva Kendram (RSK)	Pydi Bhimavaram / Narasannapeta	Soil Health Card — complete	Free (GoAP scheme)	21 days

■ **Soil Health Card Scheme:** Apply at your nearest Rythu Seva Kendram (RSK) in Srikakulam. GoAP provides a free Soil Health Card with N, P, K, pH, OC, S, Zn, Fe, Mn, Cu analysis and crop-specific fertiliser recommendations. Get this done before planting Napier.

SECTION 10 — ANNUAL COST SUMMARY & REVENUE PROJECTION

Complete financial picture for 40 Murrah buffaloes (Model B — recommended) in Pydi Bhimavaram, Srikakulam, FY 2026–27.

10.1 Annual Operating Cost Summary

Cost Head	Annual Cost (₹)	Monthly (₹)	Per Litre of Milk (₹)
Feed (Model B — green + concentrate)	13,56,960	1,13,080	8.74
Medicines, vaccines & minerals	1,70,100	14,175	1.10
Labour (3 workers @ ₹12,000/mo)	4,32,000	36,000	2.78
Electricity (motors, fans, lights)	72,000	6,000	0.46
Water (bore well running cost)	24,000	2,000	0.15
Artificial Insemination (40 AI/yr)	20,000	1,667	0.13
Veterinary consultation (12 visits)	18,000	1,500	0.12
Fodder cultivation inputs (2.5 ac)	11,250	938	0.07
Milk testing & quality (APDDCF)	12,000	1,000	0.08
Miscellaneous (repairs, transport)	36,000	3,000	0.23
TOTAL OPERATING COST	21,52,310	1,79,359	₹13.86/litre

10.2 Revenue Projection

Revenue Source	Quantity	Rate (₹)	Annual Revenue (₹)
Milk sales (31 lactating × 14 L/day × 305 days)	1,32,370 L	62/L (APDDCF rate)	82,06,940
Calf sales (10 calves/yr average)	10 calves	₹8,000/calf	80,000
Dry dung / biogas (40 animals)	—	Captive use	36,000 (electricity savings)

Revenue Source	Quantity	Rate (₹)	Annual Revenue (₹)
Spent animal sale (culls, 3–4/yr)	3 animals	₹40,000/animal	1,20,000
TOTAL ANNUAL REVENUE			₹84,42,940

10.3 Profit Summary

Item	Amount (₹)
Total Annual Revenue	84,42,940
Total Annual Operating Cost	(21,52,310)
NET ANNUAL PROFIT (before loan EMI)	62,90,630
Loan EMI (₹89.3L @ 7% × 9 years)	(11,16,000)
NET PROFIT AFTER EMI	₹51,74,630 (~₹51.7 Lakhs)
Return on Investment (post-subsidy)	~58% per annum
Break-even milk price	₹13.86/litre (well below ₹62 sale price)

■ At ₹62/litre (APDDCF procurement price), your farm generates ₹51.7 lakhs net profit per year after EMI — approximately ₹4.3 lakhs per month take-home. Milk price above ₹14/litre is pure profit.

SECTION 11 — RECOMMENDED MODEL & 90-DAY ACTION PLAN

11.1 Our Recommendation: Model B with Silage Upgrade in Year 2

For your farm at Pydi Bhimavaram with 40 Murrah buffaloes and ~2.5 acres of land, **Model B (Own Napier + Purchased Concentrate)** is the optimal starting point. In Year 2, after establishing the Napier plantation and cash flow, add a 60-tonne silage pit and transition to Model C for further cost reduction.

Pri o rit y	Recommendation	Benefit	Cost
★ ★ ★	Plant 2.5 acres CO-3 Napier immediately	Saves ■11,000+/month on green fodder	■35,000 one-time
★ ★ ★	Start urea-treating paddy straw (free from fields nearby)	Triples straw protein from 4% to 10%; free input	■500/month urea
★ ★ ★	Register with SDMPCS (milk co-operative) in Srikakulam	■62+/L guaranteed price; subsidised concentrates	Free registration
★ ★ ★	Contact AP Animal Husbandry VAS for free vaccine schedule	FMD, HS, BQ free; saves ■8,000/yr	Free
★ ★	Get Soil Health Card (Rythu Seva Kendram, Srikakulam)	Optimise fertiliser spend on Napier	Free
★ ★	Test bore well water (Oct–Nov)	Check fluoride/TDS — coastal risk	■500–1,200
★ ★	Install mineral lick blocks in shed	Passive mineral intake, improves milk fat	■200/block/month
★	Year 2: Construct 60T silage pit (maize kharif season)	Transition to Model C; saves ■15,500/month	■80,000
★	Year 2: Install biogas unit (40 animals = 20 m ³ /day)	Free cooking gas + electricity savings ■3,000/month	■1,50,000 (MNRE subsidy available)

11.2 90-Day Quick-Start Action Plan

Week	Action	Who	Cost
Week 1–2	Contact local VAS at Pydi Bhimavaram / Sompeta for vaccination schedule registration	Bhavani / farm manager	Free
Week 1–2	Apply for Soil Health Card at RSK, Narasannapeta	Bhavani	Free
Week 2–3	Collect paddy straw; start urea treatment (50 kg batches)	Farm labour	■500
Week 3–4	Source CO-3 Napier slips from ANGRAU or govt. farm — plant 2.5 acres	Labour + Bhavani	■35,000
Week 4	Collect water sample, submit to APSPCB Srikakulam for testing	Bhavani	■800
Week 5–6	Register with SDMPCS / APDDCF for guaranteed milk procurement	Bhavani	Free
Week 6–8	Set up DairyFlow app on mobile — enter all 40 animal records, calving dates, AI history	Bhavani / Suresh Babu	Free (system done)
Week 8–10	Implement daily health checklist using DairyFlow Health module	Priya Devi (Vet)	Free
Month 3	Review milk yield data on DairyFlow dashboard; optimise ration based on actual performance	Bhavani + Ravi Kumar	Feed adjustment only

11.3 Government Schemes Available to You in Srikakulam

Scheme	Benefit	Contact
NABARD DEDS (already approved)	■44.7L subsidy — already utilised for shed construction	NABARD Regional Office, Vizag
AP NTR Pashu Seva Scheme	Free veterinary services, medicines at VSU camps	AP Animal Husbandry Dept, Srikakulam
MNRE Biogas Programme	25–50% subsidy on biogas plant (central + state)	NEDCAP / DRDA Srikakulam

Scheme	Benefit	Contact
Rythu Bandhu (if eligible)	■5,000/acre/season for fodder land	Agriculture Dept, Srikakulam
PM Kisan Samman Nidhi	■6,000/yr direct benefit	CSC / bank — register if not already
APDDCF Milk Procurement	■62/L + bonus, chilling facility, quality premium	SDMPCS Srikakulam town
KCC (Kisan Credit Card)	Working capital loan at 4% interest for feed purchase	SBI / Andhra Bank, Srikakulam

SUMMARY: Your 40-buffalo Murrah farm in Pydi Bhimavaram is extremely viable.

At ■62/litre milk price:

- Annual revenue: **■84.4 lakhs**
- Annual operating cost (Model B): **■21.5 lakhs**
- Net profit after EMI: **■51.7 lakhs/year (~■4.3 lakhs/month)**

Start with Model B (Napier + concentrate). Test your water. Get the free Soil Health Card. Register with the milk co-operative. Use your DairyFlow system to track every animal daily. Upgrade to silage in Year 2.